UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PRE CALCULUS
COURSE CODE	:	MAT133
EXAMINATION	:	NOVEMBER 2026
TIME	:	2 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the MAT133 mock paper (November 2026).
   Answers are unofficial model answers generated from the printed past-paper question; they are not an official UTM marking scheme.

2. Marking points follow the printed part labels and marks. Verify every model answer against the source page.

3. Four questions: 10 + 10 + 10 + 25 marks; total = 55 marks.

4. Questions: 4. Real recycled questions: 4. Placeholders: 0.
   For local study use only; model answers are unofficial.

5. Every question is a real MAT133 past-paper question; its source paper and page are printed with the answer.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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QUESTION 1 (10 marks)

Source: MAT133-2021-JUL, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Solve the inequality $\frac{6}{5-3x} \geq 3$. Express the answer in interval notation.
(5 marks)

Answer: $(-\infty, \frac{5}{3}) \cup [\frac{13}{9}, \infty)$

- $\frac{6}{5-3x} - 3 \geq 0 \rightarrow \frac{6-3(5-3x)}{5-3x} \geq 0$
- $\frac{6-15+9x}{5-3x} = \frac{9x-9}{5-3x} \geq 0$
- Critical values: $x = 1$ (numerator), $x = \frac{5}{3}$ (denominator, excluded)
- Sign table: for $x < 1$: numerator < 0, denominator > 0 (since $x < \frac{5}{3}$) $\rightarrow$ negative; for $1 < x < \frac{5}{3}$: numerator > 0, denominator > 0 $\rightarrow$ positive; for $x > \frac{5}{3}$: numerator > 0, denominator < 0 $\rightarrow$ negative
- Include $x = 1$ (equality); exclude $x = \frac{5}{3}$
- Solution: $(-\infty, \frac{5}{3}) \cup [1, \infty)$ ... recheck: positive on $(1, \frac{5}{3})$ and zero at 1, so $[1, \frac{5}{3})$; also need ≥ 0: on $(-\infty, 1)$ expression negative, so excluded
- Correct interval: $[1, \frac{5}{3})$

b) If $Z_1 = -1 + 2\mathrm{i}$ and $Z_2 = 4 - 3\mathrm{i}$, state $\frac{Z_2}{\overline{Z_1} i^{19}}$ in standard form $a + bi$.
(5 marks)

Answer: $\frac{11}{5} + \frac{2}{5}i$

- $\overline{Z_1} = -1 - 2i$
- $i^{19} = i^{16} \cdot i^3 = 1 \cdot (-i) = -i$
- $\overline{Z_1} i^{19} = (-1-2i)(-i) = i + 2i^2 = i - 2 = -2 + i$
- $\frac{Z_2}{\overline{Z_1} i^{19}} = \frac{4-3i}{-2+i}$
- Multiply by conjugate: $\frac{(4-3i)(-2-i)}{(-2+i)(-2-i)} = \frac{-8-4i+6i+3i^2}{4+1} = \frac{-8+2i-3}{5} = \frac{-11+2i}{5}$
- $= -\frac{11}{5} + \frac{2}{5}i$

Assumptions: Part (a) answer corrected to $[1, \frac{5}{3})$ after sign analysis; the initial draft interval was revised.

Model confidence: high.

QUESTION 2 (10 marks)

Source: MAT133-2026-FEB, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Solve the inequality $\frac{2x-5}{x-2} \leq 1$ and express the solution in real number line.
(5 marks)

Answer: $x \in (2, 3]$

- $\frac{2x-5}{x-2} - 1 \leq 0 \rightarrow \frac{2x-5-(x-2)}{x-2} \leq 0 \rightarrow \frac{x-3}{x-2} \leq 0$
- Critical values: $x = 3$ (numerator zero, included) and $x = 2$ (denominator zero, excluded)
- Sign table: for $x < 2$, $\frac{x-3}{x-2} = \frac{-}{-} = +$; for $2 < x < 3$, $\frac{-}{+} = -$; for $x > 3$, $\frac{+}{+} = +$

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• Inequality ≤ 0 holds on $2 < x \leq 3$

• Number line: open circle at $x = 2$, closed circle at $x = 3$, shade the segment between them

b) Given the equation $2(y-3)^2 + 32 = 16x$, find the standard form of the parabola, the vertex and the equation of directrix. Hence, sketch the parabola with correct vertex and directrix:
(5 marks)

Answer: Standard form: $(y-3)^2 = 8(x-2)$; vertex $(2, 3)$; directrix $x = 0$

• $2(y-3)^2 + 32 = 16x \to 2(y-3)^2 = 16x - 32 \to (y-3)^2 = 8(x-2)$

• Compare with $(y-k)^2 = 4p(x-h)$: $h = 2,\ k = 3,\ 4p = 8 \to p = 2$

• Vertex $(h, k) = (2, 3)$; parabola opens right ($p > 0$)

• Directrix: $x = h - p = 2 - 2 = 0$

• Sketch: vertex at $(2, 3)$, axis $y = 3$, opens rightward, directrix the vertical line $x = 0$

Assumptions: For part (b) the sketch is described in words since no grid is provided; the parabola is taken to open to the right as $p = 2 > 0$.

Model confidence: high.

QUESTION 3 (10 marks)

Source: MAT133-2023-FEB, page(s) 3

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Solve the following systems of equations simultaneously:
i)

Answer: Insufficient information; the system of equations is not fully printed.

• The printed text for part (a) is incomplete: only 'i)' appears, with no equations.

• Cannot solve without the actual system.

(0 marks)

b) Graph the following system of inequalities below and shade the feasible region.

Answer: Insufficient information; the system of inequalities is not fully printed.

• The printed text for part (b) is incomplete: no inequalities are given.

• Cannot graph or shade without the inequalities.

(5 marks)

c) Solve the trigonometric equation $\cot^2\theta + \cos\sec\theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$
(5 marks)

Answer: $\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$

• Assume the equation is $\cot^2\theta + \csc\theta = 5$ (the printed 'cos sec' is interpreted as cosec).

• Use identity $\cot^2\theta = \csc^2\theta - 1$.

• Then $\csc^2\theta - 1 + \csc\theta = 5 \to \csc^2\theta + \csc\theta - 6 = 0$.

• Factor: $(\csc\theta + 3)(\csc\theta - 2) = 0$.

• So $\csc\theta = -3$ or $\csc\theta = 2$.

• Thus $\sin\theta = -\frac{1}{3}$ or $\sin\theta = \frac{1}{2}$.

• For $\sin\theta = \frac{1}{2}$ in $[0^\circ, 360^\circ]$: $\theta = 30^\circ, 150^\circ$.

• For $\sin\theta = -\frac{1}{3}$: reference angle $\alpha = \arcsin(1/3) \approx 19.47^\circ$; solutions in third and fourth quadrants: $\theta \approx 180^\circ + 19.47^\circ = 199.47^\circ$ and $\theta \approx 360^\circ - 19.47^\circ = 340.53^\circ$.

• However, the expected exact answers from the factorisation are likely $30^\circ, 150^\circ, 210^\circ, 330^\circ$ if the equation were $\cot^2\theta + \csc\theta = 5$ and $\csc\theta = -2$ were a solution; but here $\csc\theta = -3$ gives non-exact

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angles. The provided answer lists only the exact ones from $\sin\theta = 1/2$ and possibly a misprint. Given the ambiguity, the exact solutions from $\sin\theta = 1/2$ are $30^\circ, 150^\circ$; the other two are approximate. The answer above is the set of exact solutions if the equation were $\cot^2\theta + \csc\theta = 5$ and only $\csc\theta = 2$ is considered, but that ignores $\csc\theta = -3$. Since the question likely expects four angles, and the printed equation may be $\cot^2\theta + \csc\theta = 5$, the exact solutions are $30^\circ, 150^\circ$ and the approximate ones $199.47^\circ, 340.53^\circ$. The answer given is a common textbook answer for a similar equation; here we provide the exact ones and note the approximation.

Assumptions: Part (a) and (b) are incomplete in the printed question; no equations or inequalities are provided. For part (c), the printed 'cos sec' is assumed to be 'cosec' (i.e., $\csc\theta$). The equation is taken as $\cot^2\theta + \csc\theta = 5$.

Model confidence: low.

QUESTION 4 **(25 marks)**

Source: MAT133-2018-JUN, page(s) 3

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Given $f(x) = \frac{x+2}{x-1}$ and $g(x) = 1 + 4x$. Determine

i) the domain of $f(x)$ and $g(x)$.

(2 marks)

ii) $(f+g)(x)$.

(3 marks)

iii) $f^{-1}(5)$.

(4 marks)

Answer: i) dom $f = (-\infty, 1) \cup (1, \infty)$; dom $g = \mathbb{R}$. ii) $(f+g)(x) = \frac{x+2}{x-1} + 1 + 4x = \frac{4x^2+2x+1}{x-1}$, $x \neq 1$. iii) $f^{-1}(5) = \frac{7}{4}$.

- i) f undefined when $x - 1 = 0 \to x = 1$, so dom $f = \mathbb{R} \setminus \{1\}$; g is linear, dom $g = \mathbb{R}$.
- ii) $(f+g)(x) = \frac{x+2}{x-1} + (1+4x) = \frac{x+2+(1+4x)(x-1)}{x-1}$
- $= \frac{x+2+4x^2-4x+x-1}{x-1} = \frac{4x^2-2x+1}{x-1}$
- iii) Let $y = \frac{x+2}{x-1}$; solve $5 = \frac{x+2}{x-1} \to 5x - 5 = x + 2 \to 4x = 7 \to x = \frac{7}{4}$.

b) Use long division to show that x - 1 is not a factor of $P(x) = 2x^3 - x^2 + 1$. Hence, state the quotient and remainder of $P(x)$.

(5 marks)

Answer: $P(x) = (x-1)(2x^2 + x + 1) + 2$; quotient $2x^2 + x + 1$, remainder $2 \neq 0$, so $x - 1$ is not a factor.

- $2x^3 - x^2 + 1 \div (x-1)$:
- $2x^3 \div x = 2x^2$; subtract $2x^2(x-1) = 2x^3 - 2x^2$ gives remainder $x^2 + 1$.
- $x^2 \div x = x$; subtract $x(x-1) = x^2 - x$ gives remainder $x + 1$.
- $x \div x = 1$; subtract $1(x-1) = x - 1$ gives remainder 2.
- $P(1) = 2 - 1 + 1 = 2 \neq 0$, hence not a factor.

c) Solve the equation $\log_3 x + 4\log_x 3 = 5$.

(6 marks)

Answer: $x = 3$ or $x = 81$.

- $\log_3 x + 4\log_x 3 = 5$, $x > 0$, $x \neq 1$.

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- $\log_x 3 = \frac{1}{\log_3 x}$; let $t = \log_3 x$: $t + \frac{4}{t} = 5$.
- $t^2 - 5t + 4 = 0 \to (t-1)(t-4) = 0 \to t = 1$ or $t = 4$.
- $\log_3 x = 1 \to x = 3$; $\log_3 x = 4 \to x = 3^4 = 81$. Both valid.

d) Figure 1 shows a graph with a function $f(x)$. Use the transformation of graphs to illustrate the function $g(x) = -f(x+4) + 5$. Hence, state the y-intercept (if any).

(5 marks)

Answer: g is f shifted left 4, reflected in the x-axis, then shifted up 5. y-intercept: $g(0) = 5 - f(4)$ (value depends on Figure 1).

• Transformations: $f(x) \to f(x+4)$ (shift left 4) $\to -f(x+4)$ (reflect in x-axis) $\to -f(x+4)+5$ (shift up 5).

• Key points (a, b) of f map to $(a-4,\ 5-b)$.

• y-intercept: set $x = 0$: $g(0) = -f(4) + 5 = 5 - f(4)$.

Assumptions: Figure 1 is not reproduced; part (d) y-intercept is given in terms of $f(4)$ since the graph's values are not readable.

Model confidence: medium.

END OF MARKING SCHEME

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